

C Program - Day-3 Class Assignment

1. Write a C program to insert an element at a specified position.
2. Search for a given value in an array using linear search; print its index or "Not found."
3. Find the largest element in an array.
4. Count how many even and how many odd numbers are in an array.
5. Write a program to reverse an array
6. Delete a given value from an array
7. Write a C program to Swap First and Last Elements in an array by getting user input.
8. Write a C program to input two arrays and check whether they are identical.
9. Write a C program to copy all elements of one array into another array.
10. Write a C program to replace every negative number in an array with 0 and display the updated array.

MCQ questions

1. What will be the output of the following code?

```
int arr[5] = {1, 2, 3};  
printf("%d", arr[4]);
```

- a) Garbage value
- b) 3
- c) 0
- d) Compilation error

Answer:

2. Which statement correctly finds the number of elements in an array arr?

- a) length(arr)
- b) arr.length
- c) sizeof(arr) / sizeof(arr[0])
- d) count(arr)

Answer:

3. What is the output of this code?

```
int arr[3][2] = {{1, 2}, {3, 4}, {5, 6}};  
printf("%d", arr[2][1]);
```

- a) 5
- b) 6
- c) 4
- d) 2

Answer:

4. In C, local arrays declared inside a function without initialization contain:

- a) Always 0
- b) Always 1
- c) Garbage (indeterminate) values
- d) NULL

Answer:

5. What does the following code print?

```
int arr[] = {10, 20, 30, 40, 50};  
for (int i = 4; i >= 0; i--)  
    printf("%d ", arr[i]);
```

- a) 10 20 30 40 50
- b) 50 40 30 20 10
- c) Compilation error
- d) 40 30 20 10 0

Answer:

6. Which of these correctly deletes the element at index 2 from an array (shifting the rest left)?

```
int arr[5] = {10, 20, 30, 40, 50};  
int size = 5;  
a) arr[2] = 0;  
b) for(int i = 2; i < size-1; i++) arr[i] = arr[i+1]; size--;  
c) arr[2] = arr[size-1]; size--;  
d) free(arr[2]);
```

Answer:

7. What is the output?

```
int arr[4] = {5, 10, 15, 20};  
int sum = 0;  
for (int i = 0; i < 4; i++) {  
    sum += arr[i];  
    if (i == 2) break;  
}  
printf("%d", sum);
```

- a) 50
- b) 30
- c) 15
- d) 45

Answer:

8. Given this partially filled 2D array, what is the output?

```
int arr[3][3] = { {1, 2}, {3}, {4, 5, 6} };  
printf("%d %d %d", arr[0][2], arr[1][1], arr[1][2]);
```

- a) Garbage garbage garbage
- b) 0 0 0
- c) Compilation error
- d) 2 3 0

Answer:

9. What is the output of this code?

```
int arr[5] = {1, 2, 3, 4, 5};  
for (int i = 0; i < 5; i++) {  
    arr[i] = arr[4 - i];  
}  
printf("%d %d %d", arr[0], arr[2], arr[4]);
```

- a) 5 3 1
- b) 5 3 5
- c) 1 3 5
- d) 5 3 3

Answer:

10. What is the output of this code?

```
int arr[2][2][2] = {1, 2, 3, 4, 5, 6, 7, 8};  
printf("%d", arr[1][0][1]);
```

- a) 4
- b) 6
- c) 5
- d) 7

Answer:

Explain

1. Differentiate between compile-time initialization and run-time initialization of arrays, with examples of each.
2. How is the length of an array calculated in C, given that C has no built-in length function? Explain with the underlying logic.
3. Explain how a 2D array is stored in memory, and how the compiler computes the address of a specific element `arr[i][j]`.
4. Write the syntax for declaring and initializing the following:
 - (a) A 1D array
 - (b) A 2D array
 - (c) A 3D array

5. Write the output of the following programs:

(a) 1D Array

```
int a[] = {10, 20, 30, 40};  
printf("%d %d", a[1], a[3]);
```

(b) 2D Array

```
int a[2][2] = {{1, 2}, {3, 4}};  
printf("%d %d", a[0][1], a[1][0]);
```

(c) 3D Array

```
int a[2][2][2] = {  
    {{1,2},{3,4}},  
    {{5,6},{7,8}}  
};  
printf("%d %d", a[0][1][1], a[1][0][0]);
```

6. What happens if you try to access an element outside the array bounds?

7. Find the Error

```
#include <stdio.h>  
int main()  
{  
    int a[3] = {10, 20, 30};  
    printf("%d", a[3]);  
    return 0;  
}
```

Question:

1. Is there any error in this program?
2. Explain why it occurs.
3. Write the corrected statement.

8. What is wrong with this program?

```
int a[5];  
for(int i = 0; i <= 5; i++)  
{  
    scanf("%d", &a[i]);  
}
```

- Identify the mistake.
- What will happen during execution?
- Correct the program.

9. Find the Output

```
int a[] = {2,4,6,8,10};  
for(int i=0;i<5;i++)  
{  
    if(i%2==0)  
        printf("%d ",a[i]);  
}
```

```
}
```

10. What is the Output?

```
int a[] = {1,2,3,4,5};
```

```
for(int i=0;i<5;i++)
```

```
{
```

```
    a[i]=a[i]*2;
```

```
}
```

```
printf("%d",a[3]);
```